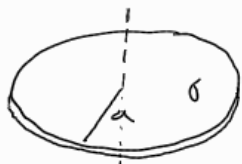


1. Električno polje nabite okrogle plošče



$E(z) = ?$

↳ na osi

a) $z \gg a$

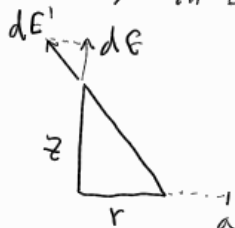
b) $z \ll a$

točkast naboj:

$$E(r) = \frac{e}{4\pi\epsilon_0 r^2}$$

kolobar:

$$dE'(z) = \frac{de}{4\pi\epsilon_0(r^2+z^2)}, \quad de = \sigma dS = \sigma \cdot 2\pi r dr$$



$$\frac{dE}{dE'} = \frac{z}{\sqrt{z^2+r^2}}$$

$$dE = \frac{z}{\sqrt{z^2+r^2}} dE' = \frac{\sigma \cdot z \cdot r dr}{2\epsilon_0 (z^2+r^2)^{3/2}}$$

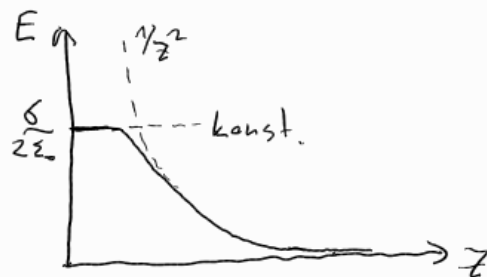
$$E(z) = \frac{\sigma z}{2\epsilon_0} \int_0^a \frac{r dr}{(z^2+r^2)^{3/2}} = \frac{\sigma z}{4\epsilon_0} \int_{z^2}^{a^2+z^2} \frac{du}{u^{3/2}} = \frac{\sigma z}{4\epsilon_0} \left[-\frac{u^{-1/2} \cdot 2}{1} \right] \Big|_{z^2}^{z^2+a^2} =$$

$$u = z^2 + r^2 \\ du = 2r dr$$

$$= -\frac{\sigma z}{2\epsilon_0} \left[\frac{1}{\sqrt{z^2+a^2}} - \frac{1}{z} \right] = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{z}{\sqrt{z^2+a^2}} \right]$$

1) $z \ll a$

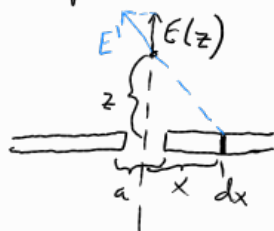
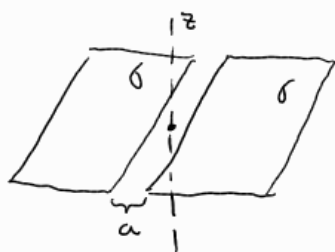
$$E = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{z}{\sqrt{z^2+a^2}} \right] \doteq \frac{\sigma}{2\epsilon_0} \quad \text{Neskončna ravna plošča}$$



2) $z \gg a$

$$E = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{z \cdot 1}{z \sqrt{1 + (\frac{a}{z})^2}} \right] = \frac{\sigma}{2\epsilon_0} \left[1 - \left(1 - \frac{1}{2} \frac{a^2}{z^2} \right) \right] = \frac{\sigma a^2}{4\epsilon_0 z^2} \frac{\pi}{\pi} = \frac{e}{4\pi\epsilon_0 z^2} \quad \text{točkast naboj}$$

2) El. polje nabite ravne plošče z rezo



$$E(z) = ?$$

Podproblem: nabiti ravni vodnik (žica)

$$\text{Gauss: } e = \epsilon_0 \oint \vec{E} \cdot d\vec{S} = \epsilon_0 E 2\pi r \cdot l \rightarrow E = \frac{e}{2\pi \epsilon_0 r l}$$



$$\frac{dE'}{dE} = \frac{\sqrt{z^2 + x^2}}{z}$$

$$dE' = \frac{de}{2\pi \epsilon_0 r l}, \quad de = \sigma dx$$

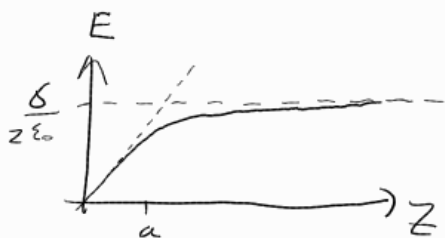
$$dE = \frac{\sigma z dx}{2\pi \epsilon_0 (x^2 + z^2)} \rightarrow E = \frac{\sigma z}{2\pi \epsilon_0} \int_{-\infty}^{\infty} \frac{dx}{x^2 + z^2} = \frac{\sigma z}{\pi \epsilon_0} \frac{1}{z} \arctg\left(\frac{x}{z}\right) \Big|_{-\infty}^{\infty} = \frac{\sigma}{\pi \epsilon_0} \left[\frac{\pi}{2} - \arctg \frac{a}{z} \right]$$

$$a) z \gg a$$

$$E = \frac{\sigma}{\pi \epsilon_0} \cdot \frac{\pi}{2} = \underline{\underline{\frac{\sigma}{2\epsilon_0}}}$$

$$b) z \ll a$$

$$E = \frac{\sigma}{\pi \epsilon_0} \left[\frac{\pi}{2} - \left(\frac{\pi}{2} - \frac{2z}{a} + \dots \right) \right] = \underline{\underline{\frac{2\sigma z}{\pi \epsilon_0 a}}}$$



3) Poissonova enačba za točkast naboj

$$\nabla^2 U = -\frac{\rho}{\epsilon_0}$$

$$\rho = e_0 \delta(\vec{r})$$

$$U = ?$$

$$\nabla^2 U = -\frac{e_0 \delta(\vec{r})}{\epsilon_0}$$

Pois. en. za točkast naboj

Fourierova transf.

$$U(\vec{r}) = \int d^3 \vec{k} U(\vec{k}) e^{i \vec{k} \cdot \vec{r}} \quad / \quad \int d^3 \vec{r} e^{-i \vec{k} \cdot \vec{r}} \delta(\vec{k} - \vec{k}') = (2\pi)^3 \delta(\vec{k} - \vec{k}')$$

$$\int d^3 \vec{r} U(\vec{r}) e^{-i \vec{k} \cdot \vec{r}} = \int d^3 \vec{k} U(\vec{k}) \int d^3 \vec{r} e^{i(\vec{k} - \vec{k}') \cdot \vec{r}} = \int d^3 \vec{k} U(\vec{k}) (2\pi)^3 \delta(\vec{k} - \vec{k}') = (2\pi)^3 U(\vec{k})$$

$$\rightarrow U(\vec{k}) = \frac{1}{(2\pi)^3} \int d^3 \vec{r} U(\vec{r}) e^{-i \vec{k} \cdot \vec{r}}$$

$$\nabla U(\vec{r}) = \int d^3k U(\vec{k}) \nabla e^{i\vec{k}\vec{r}} = \int d^3k U(\vec{k}) i\vec{k} e^{i\vec{k}\vec{r}} \rightarrow \nabla^2 U(\vec{r}) = \int d^3k (-k^2) U(\vec{k}) e^{i\vec{k}\vec{r}}$$

↑
deluje samo na $\vec{r}$

$$\nabla^2 U(\vec{r}) = -\frac{e}{\epsilon_0} \delta(\vec{r}) \xrightarrow{\text{FT}} -k^2 U(\vec{k}) = -\frac{e}{\epsilon_0} \frac{1}{(2\pi)^3} \rightarrow U(\vec{k}) = +\frac{e}{\epsilon_0} \frac{1}{(2\pi)^3} \frac{1}{k^2}$$

$$U(\vec{r}) = \int d^3k U(\vec{k}) e^{i\vec{k}\vec{r}} = \frac{e}{(2\pi)^3 \epsilon_0} \int d^3k \frac{1}{k^2} e^{i\vec{k}\vec{r}} = \frac{e}{(2\pi)^3 \epsilon_0} \int_0^\infty \int_{-1}^1 \int_0^{2\pi} k^2 d\cos\theta dk d\varphi \frac{1}{k^2} e^{i\vec{k}\vec{r}} =$$

$$\underbrace{d^3k = k^2 d\cos\theta dk d\varphi}_{(2\pi)^3 \epsilon_0} = \frac{e 2\pi}{(2\pi)^3 \epsilon_0} \int_{-1}^1 e^{ikr\cos\theta} d\cos\theta \int_0^\infty dk = \frac{e}{(2\pi)^2 \epsilon_0} \int_0^\infty \frac{1}{kr} [e^{ikr} - e^{-ikr}] dk =$$

$$= \frac{2e}{(2\pi)^2 \epsilon_0 r} \int_0^\infty \frac{\sin kr}{k} dk = \frac{e}{2\pi^2 \epsilon_0 r} \int_0^\infty \frac{\sin u}{u} du = \frac{e}{2\pi^2 \epsilon_0 r} \frac{\pi}{2} = \frac{e}{4\pi \epsilon_0 r}$$

(L) Gostota naboja v Hartenu

$$U(r) = \frac{e}{4\pi \epsilon_0} \frac{e^{-\alpha r}}{r} \left(1 + \frac{\alpha r}{2} \right); \quad \alpha = \frac{2}{r_B}$$

$$\underline{\rho(r) = ?}$$

$$\nabla^2 = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d}{dr} \right) \quad (\text{zaradi sferne simetrije})$$

$$\frac{d}{dr} U = \frac{e}{4\pi \epsilon_0} \left[\frac{-\alpha e^{-\alpha r} r - e^{-\alpha r}}{r^2} - \frac{\alpha^2 e^{-\alpha r}}{2} \right] = \frac{-e}{4\pi \epsilon_0} e^{-\alpha r} \left[\frac{\alpha}{r} + \frac{1}{r^2} + \frac{\alpha^2}{2} \right]$$

$$\nabla^2 U = \frac{1}{r^2} \frac{d}{dr} \left(-\frac{e}{4\pi \epsilon_0} e^{-\alpha r} \left[\alpha r + 1 + \frac{\alpha^2}{2} r^2 \right] \right) =$$

$$= -\frac{e}{4\pi \epsilon_0} \frac{1}{r^2} \left[-\alpha e^{-\alpha r} \left[1 + \alpha r + \frac{\alpha^2}{2} r^2 \right] + e^{-\alpha r} \left[\alpha^2 r + \alpha \right] \right] =$$

$$= \frac{e}{4\pi \epsilon_0} e^{-\alpha r} \left[\frac{\alpha}{r^2} + \frac{\alpha^2}{r} + \frac{\alpha^3}{2} - \frac{\alpha^2}{r} - \frac{\alpha}{r^2} \right] = \frac{e}{4\pi \epsilon_0} e^{-\alpha r} \frac{\alpha^3}{2} = \frac{e \alpha^3 e^{-\alpha r}}{8\pi \epsilon_0}$$

$$\rightarrow \underline{\underline{\rho_- = -\epsilon_0 \nabla^2 U = -\frac{e \alpha^3 e^{-\alpha r}}{8\pi}}}$$

$$\underline{\underline{\rho_+ = e \delta(\vec{r})}} \leftarrow \text{zaradi singularnosti } U \text{ pri } r=0$$

$$\rightarrow \boxed{\rho = \rho_+ + \rho_- = e \delta(\vec{r}) - \frac{e \alpha^3}{8\pi} e^{-\alpha r}}$$

$\nabla^2 U = 0$ skoraj povsod $\rightarrow$ Laplaceova enačba $\rightarrow$ tam, kjer ni: Robni pogoji

⑤ Ploščni kondenzator s prečnim trakom



a, U_0

$U(x, y, z) = ?$ znotraj kondenzatorja

$\rightarrow$ lahko se omejimo na x, y -ravnino, $x > 0$ (simetrije)

$$\nabla^2 U(x, y) = 0$$

robni pogoji: $U(x, 0) = 0$ $U(0, y) = U_0$

$U(x, a) = 0$ $U(\infty, y) < \infty$

Nastavek: $U = X(x)Y(y)$

$$\rightarrow X''Y + XY'' = 0$$

$$\frac{X''}{X} = -\frac{Y''}{Y} = k^2$$

zač. pog.: $X(\infty) \neq \infty$
 $\rightarrow A = 0$

$$\rightarrow X'' - k^2 X = 0$$

$$X = Ae^{kx} + Be^{-kx}$$

$$Y'' + k^2 Y = 0$$

$$Y = C \sin ky + D \cos ky$$

$$Y(0) = 0$$

$$\rightarrow \boxed{D = 0}$$

$$U(x, a) = F e^{-kx} \sin ka = 0$$

$$\rightarrow ka = n\pi$$

$$\boxed{k = \frac{n\pi}{a}} \quad n \in \mathbb{N}$$

$$\Rightarrow U(x, y) = \sum_{n=1}^{\infty} F_n \sin \frac{n\pi}{a} y e^{-\frac{n\pi}{a} x}$$

$$U(0, y) = U_0$$

$$\rightarrow \sum_{n=1}^{\infty} F_n \sin \left(\frac{n\pi}{a} y \right) = U_0 \quad \left/ \int_0^a dy \sin \left(\frac{m\pi}{a} y \right) \right.$$

(skalarni produkt)

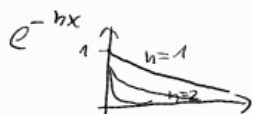
$$\sum F_n \int \sin \frac{n\pi}{a} y \sin \frac{m\pi}{a} y dy = \sum F_n \delta_{mn} \int_0^a \sin^2 \frac{n\pi}{a} y dy = \sum_{n=1}^{\infty} F_n \delta_{mn} \frac{a}{2} = F_m \frac{a}{2}$$

$$U_0 \int_0^a dy \sin \left(\frac{m\pi}{a} y \right) = \frac{U_0 a}{m\pi} \int_0^{m\pi} \sin u du = -\frac{U_0 a}{m\pi} \cos u \Big|_0^{m\pi} = \frac{U_0 a}{m\pi} (1 - \cos m\pi)$$

$$\rightarrow F_m = \frac{2U_0}{m\pi} (1 - \overbrace{\cos m\pi}^{(-1)^m})$$

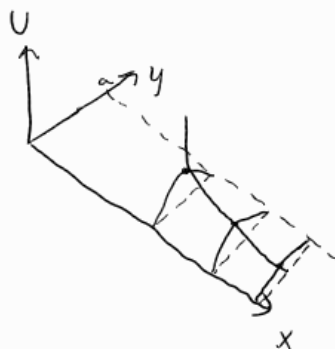
$$\rightarrow U(x, y) = \sum_{n=1}^{\infty} \frac{2U_0}{n\pi} (1 - (-1)^n) e^{-\frac{n\pi}{a} x} \sin \left(\frac{n\pi}{a} y \right)$$

a) $x \gg a$



$x \gg a \Rightarrow$ obdržimo samo $n=1$:

$$U(x \gg a, y) = \frac{4U_0}{\pi} \sin\left(\frac{\pi}{a} y\right) e^{-\frac{\pi}{a} x}$$



b) $y = \frac{a}{2}$

$$U(x, \frac{a}{2}) = \sum_{n=1}^{\infty} \frac{2U_0}{\pi n} (1 - (-1)^n) \sin\left(\frac{n\pi}{2}\right) e^{-\frac{n\pi}{a} x}$$

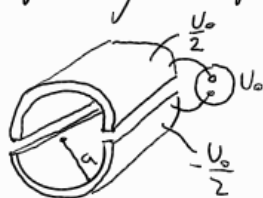
$$E_x = -\frac{\partial U}{\partial x} = -\sum_{n=1}^{\infty} \frac{2U_0}{\pi n} (1 - (-1)^n) \sin\left(\frac{n\pi}{2}\right) \left(-\frac{n\pi}{a}\right) e^{-\frac{n\pi}{a} x} =$$

$$= \sum_{n=1}^{\infty} \frac{2U_0}{a} (1 - (-1)^n) \sin\left(\frac{n\pi}{2}\right) e^{-\frac{n\pi}{a} x} = \frac{4U_0}{a} \left[e^{-\frac{\pi x}{a}} - e^{-\frac{3\pi x}{a}} + \dots \right] =$$

$$= \frac{4U_0}{a} e^{-\frac{\pi x}{a}} \left[1 - e^{-\frac{2\pi x}{a}} + \left(e^{-\frac{2\pi x}{a}}\right)^2 - \dots \right] =$$

$$\left(\begin{array}{c|c|c|c|c|c} n & 1 & 2 & 3 & 4 & 5 \\ \hline \sin & 1 & 0 & -1 & 0 & 1 \end{array} \right) = \frac{4U_0}{a} e^{-\frac{\pi x}{a}} \frac{1}{1 + e^{-\frac{2\pi x}{a}}} = \frac{4U_0}{a} \frac{1}{e^{\frac{\pi x}{a}} + e^{-\frac{\pi x}{a}}} = \frac{2U_0}{a} \frac{1}{\cosh\left(\frac{\pi x}{a}\right)}$$

⑥ Prepolovljena prevodna cev



$U(r, \varphi) = ?$
 $r < a$

$$\nabla^2 U(r, \varphi) = 0 \text{ znotraj}$$

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{d}{dr} \right) + \frac{1}{r^2} \frac{d^2}{d\varphi^2} = \nabla^2$$

$$\text{Nastavek: } U = R(r) \Phi(\varphi)$$

$$\frac{1}{r} (r R')' \Phi + \frac{1}{r^2} R \Phi'' = 0 \quad / \cdot \frac{r^2}{R \Phi}$$

$$\frac{r(R' + r R'')}{R} = - \frac{\Phi''}{\Phi} = m^2 = \text{konst.}$$

Robni pogoji:

$$U(a, \varphi) = \begin{cases} U_0/2; & 0 < \varphi < \pi \\ -U_0/2; & \pi < \varphi < 2\pi \end{cases}$$

$$U(r \rightarrow 0, \varphi) \neq \infty$$

$$\rightarrow \Phi(\varphi) = A \sin m\varphi + B \cos m\varphi, \quad m \in \mathbb{N} \text{ (ker rabimo periodičnost)}$$

$$R(r) = C r^m + D r^{-m} \text{ (povedal asistent)}$$

$$\rightarrow U(r, \varphi) = \sum_{m=1}^{\infty} (A_m \sin m\varphi + B_m \cos m\varphi) (C_m r^m + D_m r^{-m})$$

(ker je $U(\varphi)$ lična)

$$\rightarrow U(r, \varphi) = \sum_{m=1}^{\infty} F_m \sin m\varphi r^m$$

$$U(a, \varphi) = \sum_{m=1}^{\infty} F_m \sin m\varphi a^m = \begin{cases} U_0/2; & 0 < \varphi < \pi \\ -U_0/2; & \text{sicer} \end{cases} \quad / \cdot \int_0^{2\pi} d\varphi \sin n\varphi$$

$$\rightarrow \sum_{n=1}^{\infty} F_n \delta_{nn} a^n = F_n a^n \pi = \int_0^{\pi} \frac{U_0}{2} \sinh n \varphi d\varphi - \int_{\pi}^{2\pi} \frac{U_0}{2} \sinh n \varphi d\varphi = -\frac{U_0}{2n} \cosh n \varphi \Big|_0^{\pi} + \frac{U_0}{2n} \cosh n \varphi \Big|_{\pi}^{2\pi} = \frac{U_0}{n} \frac{(1-(-1)^n)}{2}$$

$$\rightarrow F_n = \frac{2U_0}{na^n \pi} (1-(-1)^n)$$

$$\rightarrow U(r, \varphi) = \sum_{n=0}^{\infty} \frac{U_0}{\pi} \left(\frac{r}{a}\right)^n \frac{1-(-1)^n}{n} \sinh n \varphi$$

Vodoravna ravnina skozi os cevi:

$$E_{\varphi} = -\frac{1}{r} \frac{dU}{d\varphi} \Big|_{\varphi=0} = \dots = -\frac{1}{r} \frac{U_0}{\pi} \sum_{n=1}^{\infty} \left(\frac{r}{a}\right)^n (1-(-1)^n) = -\frac{1}{r} \frac{U_0}{\pi} \left(2\frac{r}{a} + 2\left(\frac{r}{a}\right)^3 + 2\left(\frac{r}{a}\right)^5 + \dots\right) =$$

$$= -\frac{2}{a} \frac{U_0}{\pi} \underbrace{\left[1 + \left(\frac{r}{a}\right)^2 + \left(\frac{r}{a}\right)^4 + \dots\right]}_{\frac{1}{1-\left(\frac{r}{a}\right)^2}} = \frac{2}{a} \frac{U_0}{\pi} \underbrace{\frac{1}{1-\left(\frac{r}{a}\right)^2}}_{\text{pol v } r=a}$$

Navpična ravnina skozi os cevi:

$$E_r = -\frac{dU}{dr} \Big|_{\varphi=\frac{\pi}{2}} = \dots = -\frac{2U_0}{\pi a} \frac{1}{1+\left(\frac{r}{a}\right)^2}$$

7. Prevodna krogla v hom. el. polju



a) $U(r, \theta, \varphi) = ?$
osna simetrija

b) $\sigma(\theta) = ?$ inducirani naboji na površini

c) $\rho_e = ?$

$$U(r, \theta) = \sum_{l=0}^{\infty} (A_l r^l + B_l r^{-(l+1)}) \overset{\text{leg. polinom}}{P_l}(\cos \theta)$$

$$\int_{-1}^1 P_l(x) P_{l'}(x) dx = \frac{2\delta_{ll'}}{2l+1}$$

Robni pogoji:

1) $U(r \rightarrow \infty, \theta) = -E_0 z$ (homogeno polje v ∞)
 $\rightarrow r \cos \theta$

2) $U(a, \theta) = 0$ (ekvipot. ploskev, izbrano potencial 0)

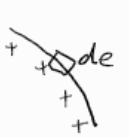
$$U(r \rightarrow \infty, \theta) = -E_0 r P_1(\cos \theta) = \sum_{l=0}^{\infty} A_l r^l P_l(\cos \theta) = A_1 r P_1(\cos \theta)$$

$$\rightarrow A_1 = -E_0, A_{l \neq 1} = 0$$

$$U(r, \theta) = -E_0 r \cos \theta + \sum_{l=0}^{\infty} B_l r^{-(l+1)} P_l(\cos \theta)$$

$$0 = U(a, \theta) = E_0 a \cos \theta + B_1 a^{-2} P_1(\cos \theta) \rightarrow B_1 = E_0 a^3, B_{l \neq 1} = 0$$

$$\rightarrow U(r, \theta) = \underbrace{-E_0 r \cos \theta}_{\text{potencial hom. el. polja}} + \underbrace{E_0 a^3 \frac{\cos \theta}{r^2}}_{\text{oblika točkastega dipola}}$$

b)  $\sigma = \epsilon_0 E_r = -\epsilon_0 \frac{\partial U}{\partial r} \Big|_{r=a} = \epsilon_0 E_0 \cos \theta + \epsilon_0 E_0 a^3 \frac{\cos \theta}{a^3} (2) =$

$$\boxed{= 3 \epsilon_0 E_0 \cos \theta}$$

c) Neformalno:

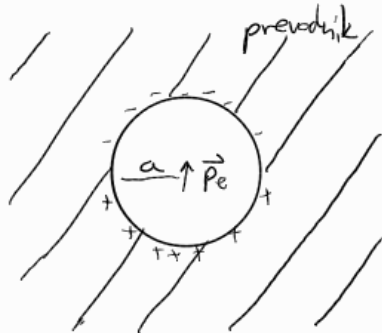
$$U_{0IP} = \frac{p_e \cos \theta}{4\pi \epsilon_0 r^2} \rightarrow \frac{p_e}{4\pi \epsilon_0} = E_0 a^3 \rightarrow \boxed{p_e = E_0 a^3 4\pi \epsilon_0}$$

Formalno:

$$\vec{p}_e = \int \vec{r} de \rightarrow \vec{p}_{ez} = \int z de = \int a \cos \theta \sigma dS = \int a \cos \theta 3 \epsilon_0 E_0 \cos \theta d \cos \theta d\varphi a^2 =$$

$$= 3a^3 \epsilon_0 E_0 \int_{-1}^1 \int_0^{2\pi} \cos^2 \theta d(\cos \theta) d\varphi = 3a^3 \epsilon_0 E_0 \cdot 2\pi \left. \frac{\cos^3 \theta}{3} \right|_{-1}^1 = \boxed{4\pi \epsilon_0 E_0 a^3}$$

8. Točkast dipol v središču krogelne votline prevodnika



a) $U(r, \theta) = ?$ znotraj

b) $\sigma(\theta) = ?$

c) $p_{e, \text{inducirani}} = ?$

R.P.:

1) $U(r \rightarrow 0, \theta) = \frac{p_e \cos \theta}{4\pi \epsilon_0 r^2}$

2) $U(a, \theta) = 0$

$$U(r \rightarrow 0, \theta) = \sum_{l=0}^{\infty} B_l r^{-(l+1)} P_l(\cos \theta) = B_1 r^{-2} \cos \theta = \frac{p_e \cos \theta}{4\pi \epsilon_0 r^2}$$

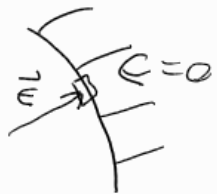
$$\rightarrow \underline{B_1 = \frac{p_e}{4\pi \epsilon_0}}, \quad B_{l \neq 1} = 0$$

$$0 = U(a, \theta) = \sum_{l=0}^{\infty} A_l a^l P_l(\cos \theta) + \frac{p_e}{4\pi \epsilon_0} a^{-2} \cos \theta = A_1 a \cos \theta + \frac{p_e}{4\pi \epsilon_0} \frac{\cos \theta}{a^2}$$

$$\rightarrow A_1 = -\frac{p_e}{4\pi \epsilon_0 a^3}, \quad A_{l \neq 1} = 0$$

$\rightarrow U(r, \theta) = \frac{p_e}{4\pi \epsilon_0} \cos \theta \left[\underbrace{-\frac{r}{a^3}}_{\text{prispevek ind. nabojev-homogeno polje, } E = \frac{p_e}{4\pi \epsilon_0 a^3}} + \underbrace{\frac{1}{r^2}}_{\text{prispevek dipola}} \right]$

b)



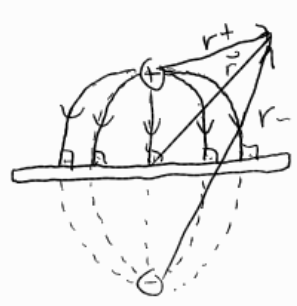
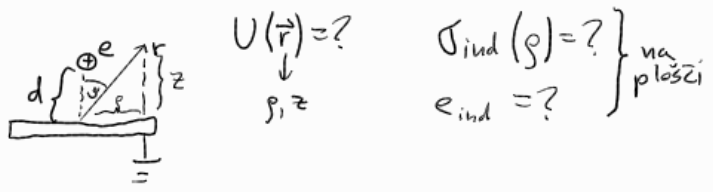
$$E_{\perp} = E_r = - \left. \frac{\partial U}{\partial r} \right|_{r=a} = \frac{3 \rho_e \cos \theta}{4 \pi \epsilon_0 a^3}$$

$$\sigma(\theta) = - \epsilon_0 E_{\perp} = \boxed{\frac{3 \rho_e \cos \theta}{4 \pi a^3}}$$

c)

$$\rho_{e, \text{ind}, z} = \dots = -\rho_e$$

9) Točkast naboj nad prevodno ploščo



polje zgoraj bo enako, če si spodaj na razdalji d neg. naboj

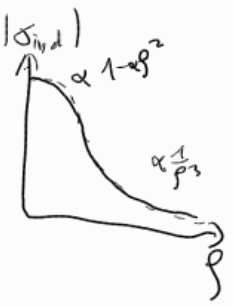
$$U(\vec{r}) = \frac{e}{4 \pi \epsilon_0} \left(\frac{1}{r^+} - \frac{1}{r^-} \right) = \boxed{\frac{e}{4 \pi \epsilon_0} \left[\frac{1}{\sqrt{(z-d)^2 + \rho^2}} - \frac{1}{\sqrt{(z+d)^2 + \rho^2}} \right]}$$

$$\sigma_{\text{ind}} = \epsilon_0 E_{\perp} \Big|_0$$

pravokotna komp. polja na površini

$$\vec{E} = -\nabla U$$

$$\vec{E}_{\perp} \Big|_0 = - \left. \frac{\partial U}{\partial z} \right|_{z=0} = - \frac{e}{4 \pi \epsilon_0} \left[- \frac{1}{2} \frac{2(z-d)}{((z-d)^2 + \rho^2)^{3/2}} + \frac{1}{2} \frac{2(z+d)}{((z+d)^2 + \rho^2)^{3/2}} \right] \Big|_{z=0} =$$

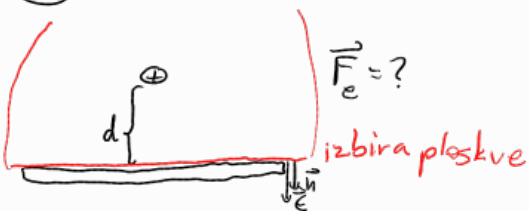


$$= - \frac{e}{4 \pi \epsilon_0} \left[d(d^2 + \rho^2)^{-3/2} + d(d^2 + \rho^2)^{-3/2} \right] = \frac{e}{2 \pi \epsilon_0} d(d^2 + \rho^2)^{-3/2}$$

$$\boxed{\sigma_{\text{ind}}(\rho) = - \frac{e}{2 \pi} d(d^2 + \rho^2)^{-3/2}}$$

$$e_{\text{ind}} = \int \sigma_{\text{ind}} dS = - \frac{ed}{2 \pi} \int_0^{\infty} \frac{2 \pi \rho d\rho}{(d^2 + \rho^2)^{3/2}} = -ed \int_{d^2}^{\infty} \frac{1}{2} \frac{du}{u^{3/2}} = ed u^{-1/2} \Big|_{d^2}^{\infty} = -e$$

10) El. sila na točkast naboj nad prevodno ploščo

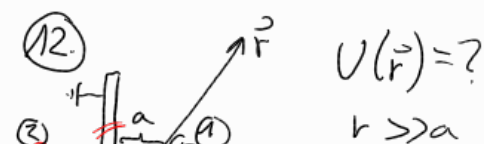


$$\vec{F}_e = \epsilon_0 \oint \left[\underbrace{\vec{E}(\vec{E} \cdot \vec{n})}_{(\vec{E} \otimes \vec{E}) \cdot \vec{n}} - \underbrace{\frac{1}{2} E^2 \vec{n}}_{-\frac{1}{2} E^2 I \vec{n}} \right] dS = \epsilon_0 \oint (\vec{E} \otimes \vec{E} - \frac{1}{2} E^2 I) \vec{n} dS$$

$$\vec{F}_{e1} \sim \epsilon_0 E^2 S \sim \epsilon_0 \left(\frac{1}{r^3} \right)^2 2\pi r^2 \sim \frac{1}{r^4} \xrightarrow{r \rightarrow \infty} 0$$

$$\vec{E} \parallel \vec{n} \rightarrow \vec{E} \cdot \vec{n} = E \rightarrow \vec{E}(\vec{E} \cdot \vec{n}) = E^2 \vec{n}$$

$$\vec{F}_{e2} = \frac{\epsilon_0}{2} \int_0^\infty \left(\frac{e a}{2\pi \epsilon_0} \right)^2 \frac{1}{(d^2 + \rho^2)^{3/2}} 2\pi \rho d\rho \hat{e}_z = \dots = -\frac{e^2}{16\pi \epsilon_0 d^2} \hat{e}_z$$



→ Multipolni razvoj: $U(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left[\frac{e_{tot}}{r} + \frac{1}{r^3} \sum_{i=\{x,y,z\}} p_i r_i + \frac{1}{r^5} \frac{1}{2} \sum_{i=\{x,y,z\}} Q_{ij} r_i r_j + \dots \right]$

$$Q_{ij} = \int d^3\vec{r} \rho(\vec{r}) (3r_i r_j - \delta_{ij} r^2) \quad \text{tr} Q = 0$$

$$e_{tot} = 0$$

$$\vec{p} = 0 \quad (\uparrow \downarrow)$$

$$\rightarrow U(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{1}{r^5} \frac{1}{2} \sum Q_{ij} r_i r_j$$

$$Q_{ij} \xrightarrow{\text{diskretno}} \boxed{Q_{ij} = \sum_{k=1}^4 e_k (3r_{ik} r_{jk} - \delta_{ij} r_k^2)} \quad \text{simetrično}$$

↑
po nabojih

$$Q_{xx} = e \underbrace{[3aa - 1(a^2 + a^2)]}_1 - e \underbrace{[3(-a)(-a) - 1(a^2 + a^2)]}_2 + e [3(-a)(-a) - 1(a^2 + a^2)] - e [3a \cdot a - 1(a^2 + a^2)] = 0$$

$$Q_{yy} = Q_{zz} = 0 \quad (\text{isto kot zgornaj})$$

$$Q_{xy} = Q_{yx} = e [3a^2 - 0] - e [-3a^2 - 0] + e [3a^2 - 0] - e [-3a^2 - 0] = 12ea^2$$

$$Q_{xz} = Q_{zx} = 0 \quad (r_z = 0)$$

$$Q_{yz} = Q_{zy} = 0$$

$$\rightarrow U(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{1}{r^5} \frac{1}{2} [Q_{xy} xy + Q_{yx} xy] = \frac{1}{4\pi\epsilon_0} \frac{1}{r^5} 12ea^2 xy = \frac{3ea^2}{\pi\epsilon_0} \frac{xy}{r^5}$$

